Lab 5: Great Salt Lake Boundary Change Analysis

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Introduction:

The climate is changing and so is the landscape of different locations. Some areas are drying up, while other areas are flooding. The Great Salt Lake in Utah is no exception. It is one of the largest saltwater lakes in the Western Hemisphere with an average depth of only 16 feet[1]. This means the area of the lake can change substantially because of the low depth. The lake is landlocked and has no outlet to the outside. Also, as the name suggests, the lake is high in salinity, which makes it not habitable for almost all species except just a few including brine shrimp, brine flies, nematodes, and different algae[1]. It is estimated that the lake contributes about $1.3 billion annually to Utah's economy[1]. In this report, I will focus on the Great Salt Lake in Utah and discuss the impact climate change has had on the size of the lake since 1889.

Data and Methods:

I began by downloading the 1889 historical image of Utah from David Rumsey[2] collection into a report file. I then imported the historical image into a new map on ArcGIS Pro with a base map of National Geographic World Map. Since the image was not a spatial layer, it did not initially show up on display, but it was still there. Before starting the georeferencing, I changed the Resampling Type to Cubic from the Raster Layer tab. Then I started the georeferencing process by going into Imagery tab and clicking on Georeference, which unlocked the Georeference tab on the top of the ribbon. Then I moved my screen to the Utah area on the base map and clicked on Fit to Display on the Georeference tab to move and fit the historical map onto my screen area.

I then clicked on Add Control Points from the Georeference tab to scale the map. I first clicked on the historical map then to where that point matched on the modern base map. I matched all the corners of Utah except the southeastern corner. The southeastern corner made the residuals very large, and it was outside the area I was concerned about, so I excluded that point. Doing this moved and scaled it to the right level for the local area I was concerned about. I then made sure to save the scaled historical map by clicking on Save as New from the ribbon with the Output Format as TIFF. Then, the georeferenced TIFF automatically got added to my map. I ended the georeferecning process by clicking on the Close Georeferencing button from the Georeference tab. I then deleted the original imagery later from my map and saved the project.

I then started the process of digitizing the Great Salt Lake based on the historical map. On a new map, I transferred the georeferenced TIFF. I clicked on the View tab then on Catalog pane. I right clicked on my report file and selected Shapefile from New. This opened the Create Feature Class window. From there, the Future Class Location was set to the report folder and Feature Class

[1] Wikipedia. " Great Salt Lake" Accessed February 13, 2025. https://en.wikipedia.org/wiki/Great_Salt_Lake
[2] David Rumsey. “Utah 1889” Accessed February 13, 2025. https://davidrumsey.com

Name to Great Salt Lake with Geometry Type as Polygon and Coordinate System of base map layer. Then clicked on run to create the shapefile.

From the Edit tab, I clicked on the Create Feature pane of the Great Salt Lake with polygon selected. I then used lines to trace the outline of the Great Salt Lake. In doing that I made sure not to include the islands shown on the lake by tracing around it. I then saved all my work from the Edit tab and closed the Create Feature window. I opened the Attribute Table for the Great Salt Lake shapefile and added a new column for Area. I then calculate the area by right clicking on the Area column and then on Calculate Geometry. The Input Feature was Great Salt Lake, Field as area with Property as Area with units as square statute miles. The coordinate system was WGS 1984 Web Mercator Auxiliary Sphere.

I converted the Great Salt Lake layer to KML file to view it in Google Earth[3] to compare the historical extent of the Great Salt Lake. I first changed the shapefile fill to hollow with a 2 point border. Then saved it to my folder as Layer File by right clicking on the layer from the Contents window. Then using the conversion tools on ArcToolbox, I selected Layer to KML. And selected the input file as the one I saved. I then opened the KML file from File Explorer on Google Earth Pro. Which showed the comparison to the most recent satellite imagery from late 2020.

I followed the similar process to create the perimeter of the Great Salt Lake using the 2020 world imagery basemap. And also to import the layer to KML so that it could be viewed on Google Earth to compare with the 1889 Area of Great Salt Lake to the 2020 Area of Great Salt Lake. On Google Earth, I made a few edits with the coloring by right clicking on the file and selecting Properties then Style, Color section. For 1889, I made the lines color to be red and area to be filled with red with 40% opacity. For 2020, I made the line color to be blue and area to be filled with blue at 40% opacity.

Results:

My findings from the digitizing of the Great Salt Lake boundary from an historical 1889 image of Utah, figure 1, show that the lake was very large coving about 2,210 square miles based on my outline of the boundary. The boundary of the lake is colored yellow in 2 points. I tried to exclude the islands that were shown in the historical image to get the most accurate area amount that the lake covered back in 1889. I then opened the KML file for this on Google Earth, figure 2, to compare my boundary and to see how the lake area changed with times from 1970 to 2020 on the app. I noticed that the lake got smaller with the years and my boundary was a little off on the southern side of the Great Salt Lake. The lake is snowing a shrinking trend in its area.

In figure 3, you will find the root mean square error (RMSE) table of five of my control points during georeferencing. The highest residual was 5,522 and the lowest was 678. This high residual was caused by the discrepancy between the refence data on the modern map and the historical image. My digitalizing of the most recent satellite imagery from 2020, show that the area of the lake is about 978 square miles. Which is 1,232 square miles smaller than the area of the lake in

[3] Google Earth. (2024). "Great Salt Lake, Utah [Satellite image]". Accessed February 15, 2025. https://earth.google.com/

1889. In figure 4, I compare the historical lake size with the recent lake size in 2020 visually. It shows that the lake has shrunk by a lot and the area result for the two digitalized layers tell the exact numbers.

Discussion:

In figure 1, the yellow outline around Great Salt Lake is exactly like the actual historical image showing the lake. However, on figure 2, the southern islands or land is a little off and needs to be shifted to the west by a small amount, otherwise the northern side of the lake looks fine. The figure 3, which may have been concerning but the map was accurate to the local area of the Great Salt Lake. I originally had more than five, but it made the residuals way higher, so I excluded them. Using the Attribute Table I found the area to be about 2,210 square miles, which translates to about 572,387 hectares. My digitalized Great Salt Lake of 2020 world view map was very accurate to the actual map on Google Earth. The area I got from the Attribute Table was 978 square miles, which translates to about 253,301 hectares. I try to show the area difference between the two years in figure 4. The red shows the area of the lake in 1889 from the digitized historical map and the blue shows the area of the lake in 2020 from the world view map.

From what I saw, the lake is shrinking and soon may disappear. The lake had a net loss of about 1,232 square miles of area, which is about 319,087 hectares. That is about 55.75% net loss in the area in the 131 years. Due to recent temperature extremes, 131 years from 2020 the lake may not exist. This may impact the Utah economy as its contributes to about 1.3 billion dollars to its economy. There may not be much loss in ecological habitats since it does not have many habitats. However, it raises concerns about those that do have large number of biological habitats. The shrinking lakes of those areas may suffer damage to their ecological habitats.

Conclusion:

The Great Salt Lake in Utah has experienced a 55.75% net loss in its area. It originally covered about 2,210 square miles of area in 1889 but in 2020 it only covered about 978 square miles. In digitalizing the image, the residual may be high, but it does not impact on the local area of concern that has the points. My digitizing of the 1889 lake was a little off and it could have been due to the type of mapping used in 1889 one and the modern one. If they were both the same, the boundaries on Google Map could have been the same. The 2020 version of the lake was very actual with the boundary. Overall, with no question the Great Salt Lake has experienced a large decrease in its area, which could raise concerns about other lakes around the world.

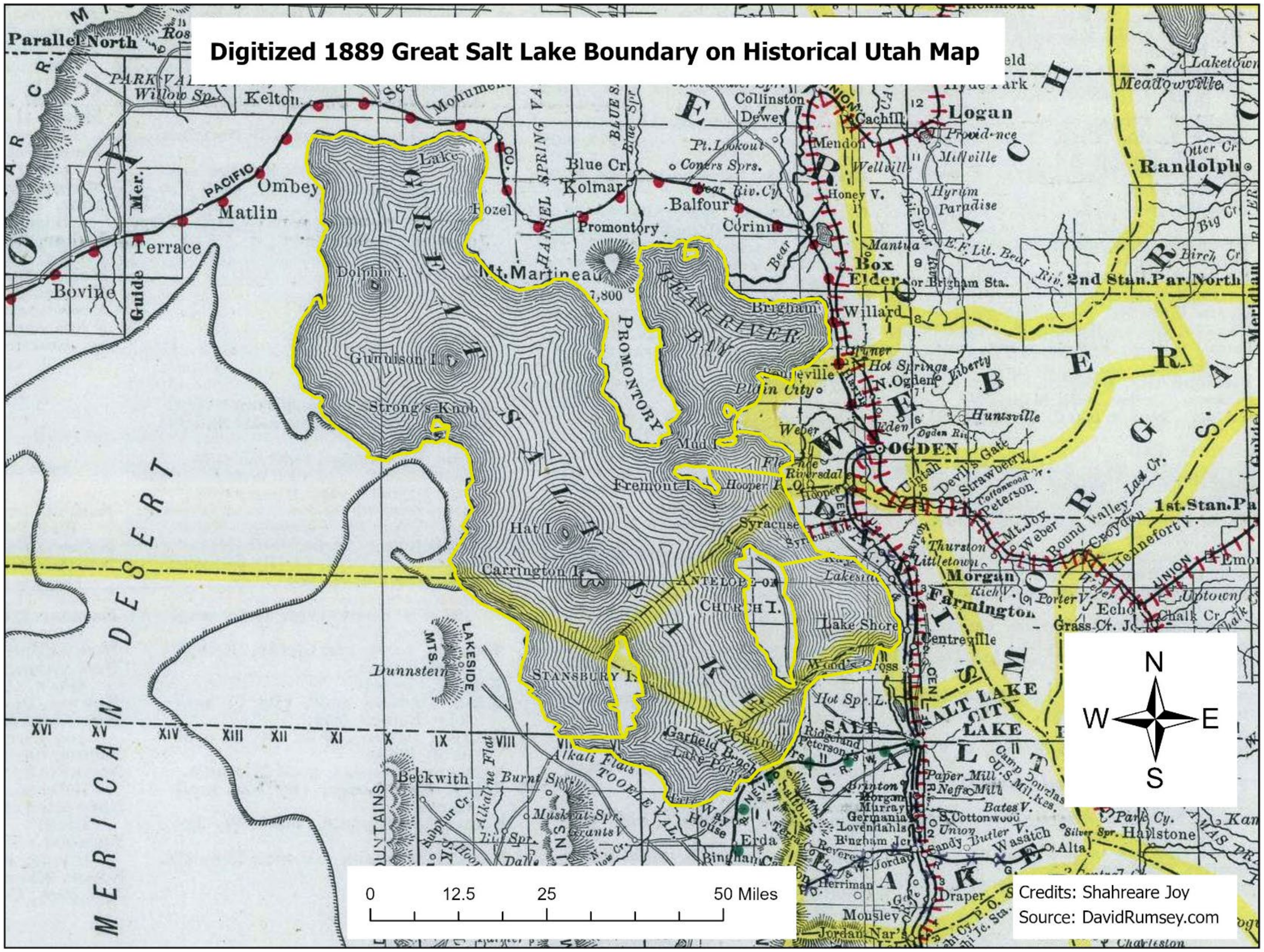

Figure 1: A digitized boundary of the Great Salt Lake overlaid in dark yellow on a historical image of Utah from 1889.

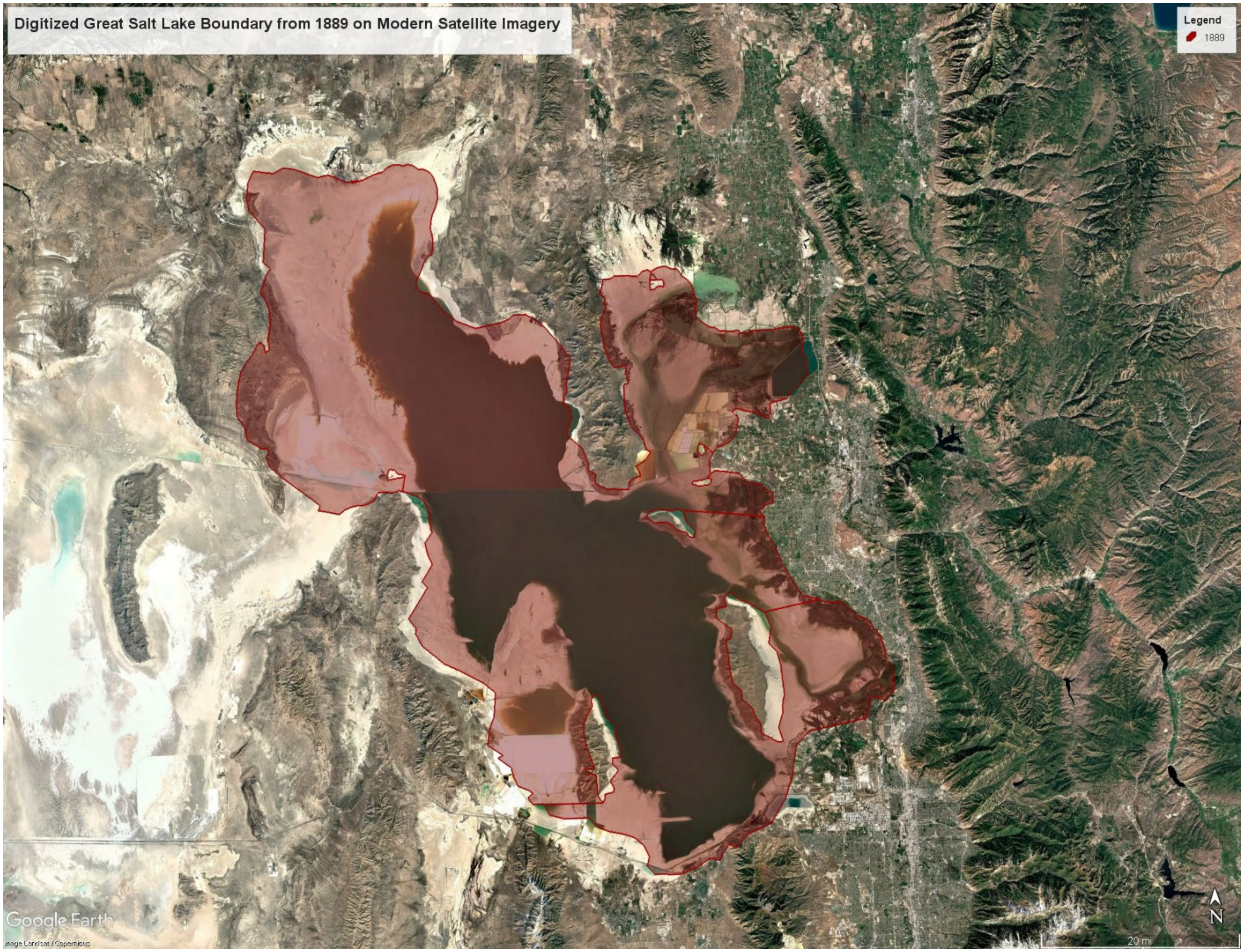

Figure 2: The digitized Great Salt Lake polygon on the modern satellite imagery (in red) on Google Earth.

1st Order Polynomial (Affine)

	Link	Source X	Source Y	Map X	Map Y	Residual X	Residual Y	Residual
☑	1	1,769.893952	-1,354.581464	-12,361,687.9557...	5,012,740.866299	-1,685.109304	787.899931	1,860.209576
☑	2	1,764.718710	-734.019081	-12,361,942.7844...	5,161,053.813759	3,444.820253	4,315.204513	5,521.573740
☑	3	378.415165	-706.264340	-12,695,087.8032...	5,159,894.325263	-1,269.062216	-2,495.935103	2,800.037669
☑	4	2,704.247777	-1,343.304105	-12,139,867.1771...	5,012,113.979041	-1,022.449750	-3,025.601924	3,193.692298
☑	5	279.269612	-3,805.574463	-12,695,988.9261...	4,439,623.196953	531.801016	418.432583	676.681718

Figure 3: Table of RMSE values for the control points used in georeferencing the 1889 Utah map onto the modern map.

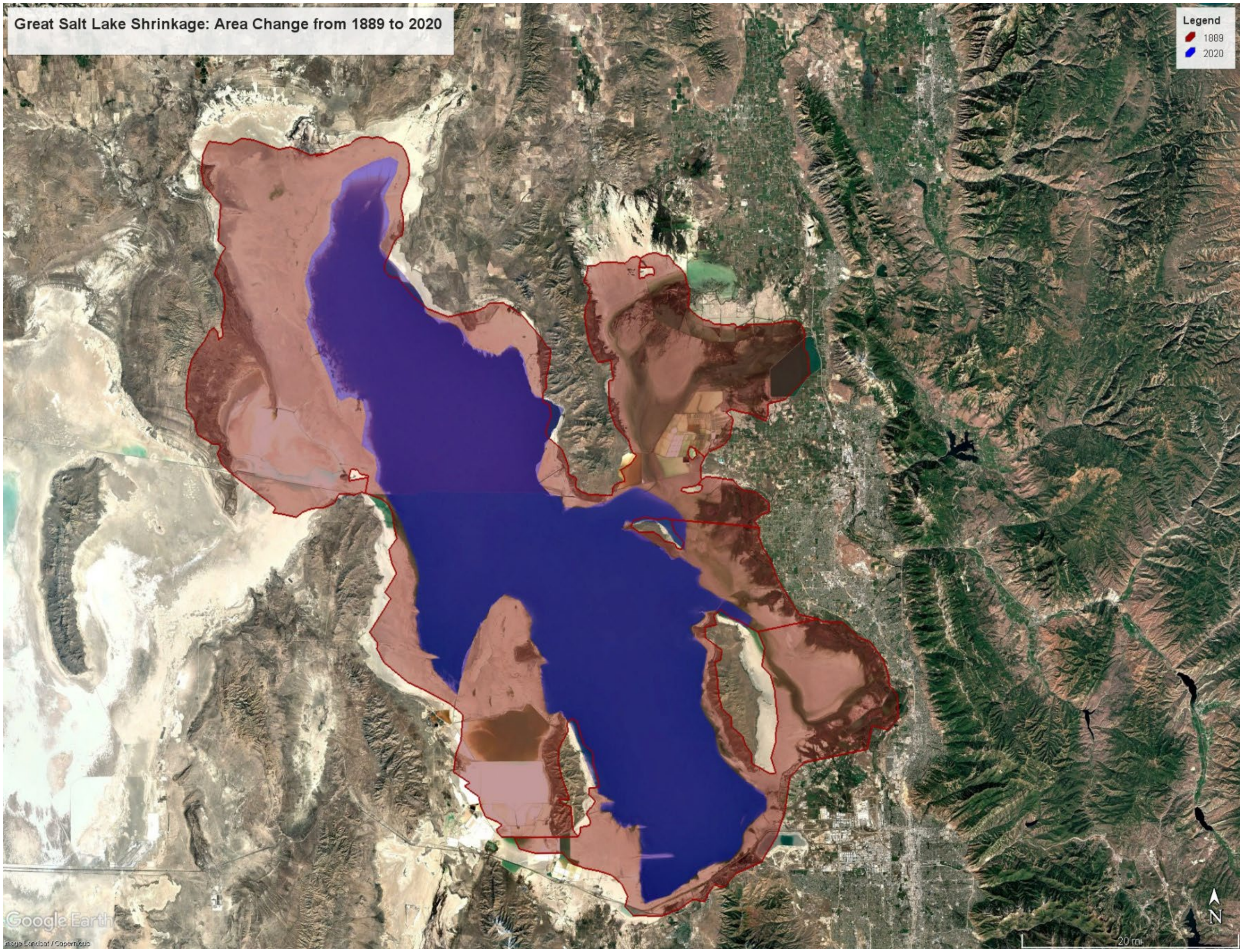

Figure 4: A satellite imagery of the Great Salt Lake from Google Earth, comparing its area in 1889 (in red) and 2020 (in blue).